Make a short drive

Name: ______________________

Date: ______________ **Period:** ______________

Today I can...

Make the robot stop after a short time. Measure how far it travels.

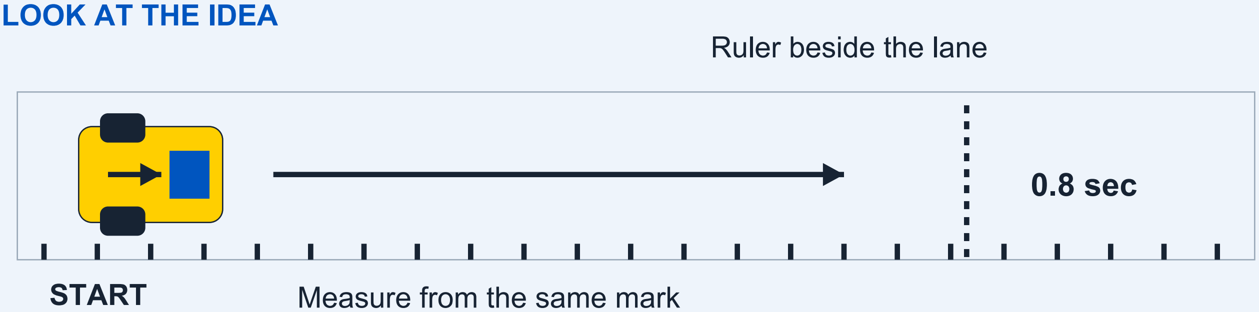
Words to know

Motor: A part that makes something move.

Bounded: Set to stop after a certain time or number of steps.

Trial: One test run.

LOOK AT THE IDEA



Gather your materials

- Checked wheeled base and connected device
- Ruler or measuring tape
- Removable floor tape
- Paper and pencil

Build a measured driving lane

Build with the power off. Tick each step when it is done.

1. Choose a flat floor space with the teacher. Keep it away from stairs and table edges. Make the lane at least twice as wide as the robot.
2. Mark a start line across the lane. Put the ruler beside the lane, with zero at the start line. Keep the ruler out of the wheel path.
3. Add a small tape mark on the side of the robot, above its main wheel axle. Use this same mark to measure every run. An axle is the rod a wheel turns on.
4. With the power off, turn each wheel gently by hand. Remove anything rubbing a wheel. Check that the sensor mount stays clear of the floor.

Sketch the setup. Label the parts you will check.

Finish the setup

5. Raise the wheels on firm supports. Try the short program below. If the wheels would send the robot backward, stop and ask the teacher to correct the motor direction.

6. Place the robot's measuring mark over the start line. Clear the lane. With the teacher's approval, run the program and wait until the wheels stop.

7. Measure from the start line to the same mark on the stopped robot. Write the distance in centimeters (cm). Return it to the start with the power off.

8. Repeat three times with the same code. Then ask the teacher to help test the Stop control during another short run. Save the lane and measuring mark.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Show where your robot starts and how you will stop it.

Code it. Predict. Try it.

Make the program

1. Start with movement stopped. Set low speed, such as 20 percent.

2. Move forward for 0.8 seconds, then stop.

3. End the program. Do not use a forever-repeat block.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

Test it fairly

Name: ______________________________

Date: ____________________ **Period:** ____________________

Change only one thing, such as speed. Keep the time, floor, load and start line the same.

A successful test

- ☐ The robot stops by itself after each short drive.
- ☐ All three distances are recorded from the same mark.
- ☐ The teacher checks the screen Stop control and power-off method.

Record every result

Run	Speed setting	Distance (cm)	Stopped by itself?
1			
2			
3			

Which result stands out?

____

Change one thing. Try again.

The one thing we changed:

We kept these things the same:

Run	Speed setting	Distance (cm)	Stopped by itself?
1			
2			
3			

Why might three runs travel slightly different distances?

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.